

D. Apes Together Strong



Time Constraints

• **java** : 3.0 Seconds



Memory Constraints

• **java** : 256MB

In the year 3034, the apes have conquered the earth and established their realm ApeGuard. The humans, led by Sergeant John Conor, aim to disrupt the apes' unity by bombing key cities. The goal is to find the minimum number of cities that need to be destroyed to disconnect the ApeGuard realm.

Input Format

1. The first line contains two integers: **N** (number of nodes/cities) and **E** (number of edges).
2. The following **E** lines contain **A** and **B** integers, representing the edges between nodes.

Output Format

The output is a single integer representing the minimum number of nodes/cities that need to be destroyed to disconnect the ApeGuard realm. Do not print extra spaces.

Constraints

$1 \leq N \leq 1000$,
 $1 \leq E \leq 1000$

Sample Cases

• Sample Input 1:

```
6 5
3 4
4 2
2 1
2 5
5 6
```

Output: **3**

• Sample Input 2:

```
4 4
1 3
1 4
3 4
2 3
```

Output: **1**